An Introduction to Higher Algebraic K-Theory

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Chapter 1

Projective Modules

The subject of K-Theory concerns with studying vector bundles on spaces. Algebraic K-Theory studies the algebraic analogue of the same, projective modules over associative rings.

1.1 Projective Modules

1.1.1 Idempotence Criteria for Projective Modules

1.1.2 Constructing Projective Modules

- Patching along localizations
- Milnor Patching

1.1.3 Classification of Projective Modules

- Projective modules ¡-¿ vector bundles -

Chapter 2

The Grothendieck Group K_0

In this chapter, we will develop the notion of the Grothendieck Group K_0 accross a variety of contexts. First, we discuss the grothendieck group of a ring, then symmetric monoidal categories.

2.1 K_0 of a ring

Let R be a (unital) ring. The interesting objects of study will be finitely generated projective modules over R. We denote by $\mathbf{P}(R)$ the set of isomorphism classes of fintely generated projective modules over R. Note the $\mathbf{P}(R)$ derives a monoid structure through $\oplus$ because direct sums yield an associative operation that preserves isomorphisms. This monoid is also abelian due to the isomorphisms $M \oplus N \cong N \oplus M$ for any R-modules.

The idea is to consider the group naturally derived by implementing inverses for all elements in this monoid.

2.1.1 Group Completion of a Monoid

Given an (abelian) monoid $(M, +)$, we can derive an (abelian) group associated to M, which we shall call $\tilde{M}$, via the following procedure:

Define an equivalence relation on $M \times M$ by

$$(a, b) \sim (c, d) \Leftrightarrow a + d + e = b + c + e \text{ for some } e \text{ in } M.$$

This indeed gives us an equivalence relation. Let $\tilde{M}$ be the set of equivalence classes of $M \times M$ under this relation. This set admits a natural (abelian) group operation defined by

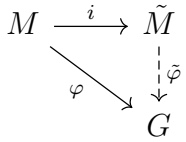
$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

This is a well-defined, abelian and associative operation on $\tilde{M}$ with the identity element $0 := [(0,0)]$ represented by any (l,l). Each element $[(a,b)]$ admits an inverse given by $[(b,a)]$ as $[(a,b)] + [(b,a)] = [(a+b, b+a)] = 0$.

A very familiar example of this construction is the abelian group $\mathbb{Z}$, which is the group completion of the abelian monoid $\mathbb{N}$ under $+$.

This construction is natural in M and follows a suitable universal property which we shall state.

Proposition 2.1 (Universal property of Group Completion)**.** We have a monoid homomorphism $i : M \to \tilde{M}$ such that for any (abelian) group G, and a monoid homomorphism $\varphi : M \to G$, we have that φ must factor uniquely through i. That is, there is a unique map $\tilde{\varphi} : \tilde{M} \to G$ such that the following diagram commutes:



Proof. The monoid homomorphism $i : M \to \tilde{M}$ is simply given by mapping a to $[(a,0)]$. This is indeed a homomorphism since $[(a,0)] + [(b,0)] = [(a+b,0)]$. Now suppose we are given $\varphi : M \to G$. Then $\tilde{\varphi} : \tilde{M} \to G$ is given by $[(a,b)] \mapsto \varphi(a) - \varphi(b)$, making the diagram commute. This map is unique since for any $a \in M$, we have $\tilde{\varphi}([(a,0)]) = \varphi(a)$, and consequently $\tilde{\varphi}([(0,a)]) = -\tilde{\varphi}([(a,0)]) = -\varphi(a)$, which altogether determines the map $\tilde{\varphi}$. □

Furthermore, any abelain group satisfying the given universal property is isomorphic to $\tilde{M}$ via a unique isomorphism.

There is an alternative (but equivalent) characterization of $\tilde{M}$ via the quotienting the free group on M by the monoidal relations.

Proposition 2.2. Let $K(M)$ be the abelian group given by taking the free abelian group $F(M)$ on the set M and quotienting by all relations of the form $[a] + [b] - [a+b]$ where $a, b \in M$ and $[x]$ represents the element in $F(M)$ corresponding to the element $x \in M$. Then $K(M) \cong \tilde{M}$.

Proof. We will show that $K(M)$ satisfies the universal property of $\tilde{M}$. Let $F(M)$ represent the free abelian group on the set M. The map $i : M \to K(M)$ on x is given by taking the equivalence class of the element x. Suppose $\varphi : M \to H$ is a monoid homomorphism to an abelian group H. Then define $\tilde{\varphi}([a])$ as $\varphi(a)$. Suppose $[a] = [b]$, then $[a] + [-b] = [0]$ so $a - b = 0$ and consequently $\varphi(a) - \varphi(b) = 0$. Thus the map $\tilde{\varphi}$ is well-defined

and clearly a group homomorphism. It is also unique since $\tilde{\varphi}$ is specified uniquely on the basis of $F(M)$. □

Example 2.1.1. Let R be a (unital) ring satisfying the *Invariant Basis Property (IBP)*. That is, $R^m \cong R^n$ implies $m = n$. Then the monoid of isomorphism classes of free R-modules with the operation $\oplus$ is isomorphic to $\mathbb{N}$, the isomorphism given by the map $R^n \mapsto n$. Consequently, its group completion is isomorphic to the abelian group $\mathbb{Z}$.

Note that all fields (and division rings) F satisfy IBP since dimension of an F-module is independent of the choice of basis. Further, any ring R with a map $R \to F$, a field or a division ring F satisfies IBP. In particular, commutative rings all satisfy IBP.

Example 2.1.2. Let X be a topological space. Then the set $[X, \mathbb{N}]$ of continuous maps $X \to \mathbb{N}$ forms an abelian monoid under pointwise addition of maps. Then the group completion of this monoid is isomorphic to the abelian group of continuous maps $X \to \mathbb{Z}$ denoted $[X, \mathbb{Z}]$. If X is quasi-compact, this is in fact a free abelian group. Notably, $[X, \mathbb{N}]$ also admits a commutative semiring structure via pointwise multiplication in addtion to the monoid operation. This makes $[X, \mathbb{Z}]$ into a commutative ring.

2.1.2 Defining $K_0(R)$

Now we can define the group $K_0(R)$ as the group completion of the monoid $\mathbf{P}(R)$ consisting of finitely generated projective R-modules with the monoid operation given by $\oplus$.

Example 2.1.3. Let F be a field or a division ring. Then all finitely generated projective F-modules are free. This gives us $K_0(F) \cong \mathbb{Z}$. More generally, for any local ring R, all finitely generated projective R-modules are free. This gives us $K_0(R) \cong \mathbb{Z}$.

Example 2.1.4. Note that finitely generated projectives over $\mathbb{Z}$ are free. Thus $K_0(\mathbb{Z}) \cong \mathbb{Z}$. More generally, for any ring R with the IBP

2.1.3 Morita Equivalence and Invariance

Suppose we have (unital) rings R and S such that we have an equivalence of the respective categories of right modules:

$$\textbf{mod-}R \leftrightharpoons \textbf{mod-}S$$

Then we say that R and S are **Morita equivalent**.

A consequence of Morita Equivalence is that R and S yield isomorphic K_0 groups.

2.2 K_0 of a Symmetric Monoidal Category

2.3 K_0 of an Exact Category

Chapter 3

Groups K_1 and K_2 of a ring

Chapter 4

Constructions of Higher K-Theory

In this chapter, we shall explore the definitions of higher K-theory. The primary goal of higher K-theory is to (functorially) define a 'K-Theory space' $K(R)$ associated to a unital ring R. We then define the higher K-groups to be $K_n(R) = \pi_n K(R)$, and verify that this description matches with the classical theory for $n = 0, 1, 2$.

A number of approaches can be taken in different contexts, and we have some results showing that in some restricted contexts, these various constructions are equivalent, i.e., yield homotopy-equivalent K-theory spaces.

Finally, this allows us to define and relate the higher K-theory of finitely generated projective modules over a ring, among other applications.

In Section §1, we first recall some homotopy-theoretic notions. In Section §2, we describe the +-construction, introduced in [] by ¡¿ and used by Quillen in [] to describe higher K-groups. Section §3 describes higher K-groups of a ring using the +-construction. In Section §4, we describe the $S^{-1}S$ construction on a symmetric monoidal category, and show its equivalence with the +-construction for $(\mathbf{P}(R), \oplus)$. In Section §4, we describe Quillen's Q-construction for exact categories. Section §5 is devoted to the $+ = Q$ theorem, which implies equivalence of the + and Q-construction for $\mathbf{P}(R)$.

4.1 Homotopy Theory Preliminaries

A based space is a topological space X along with a distinguished basepoint which we denote $*_X$. A map of based spaces $X \to Y$ is a map of the underlying topological spaces which preserves basepoints.

We can functorially associate homotopy groups π_n to a based space X, denoted $\pi_n(X)$. We assume familiarity with the definitions of homotopy and homology groups. $H_*(X; M)$ will refer to the singular homology of X with coefficients in some abelian group M, or alternatively local coefficients, i.e. M being a $\pi_1(X)$-module. When $M = \mathbb{Z}$, this is referred to as integral homology. See [] for a detailed treatment.

A CW complex (or cellular complex) X is a colimit of a filtration of n-skeleta, denoted $\emptyset = X_{-1} \subseteq X_0 \subseteq X_1 \subseteq X_2 \subseteq \cdots$ where X_0 is a discrete set of points and X_n is a pushout of X_{n-1} via a disjoint union $\cup_\alpha e_\alpha$ of attaching maps $e_\alpha : S^{n-1}_\alpha \to X_{n-1}$ and the disjoint union $\cup_\alpha j_\alpha$ of inclusions $j_\alpha : S^{n-1}_\alpha \to D^n_\alpha$. (Note S^{-1} is taken to be $\emptyset$.)

That is, $X_n = X_{n-1} \cup_{S^{n-1}_\alpha} D^n_\alpha$. The images of the interiors of the disks D^n_α in X_n are called n-cells. Note that 0-cells correspond to a set of points, 1-cells correspond to line segments, 2-cells to disks, and so on.

The colimit carries the weak topology. The homology of CW complexes is particularly tractable. See [Hatcher] for a detailed treatment.

4.1.1 Geometric Realisation of a small category

Given a small category C, there is a way to interpret C as a CW complex by the following procedure:

Definition 4.1 (Simplicial sets)**.** A simplicial set X is a contravariant functor $\mathbf{\Delta} \to \mathbf{Sets}$, where $\mathbf{\Delta}$ is the simplex category consisting of objects $\mathbf{n} = \{0, 1, \ldots, n-1\}$ and morphisms $\mathbf{i} \to \mathbf{j}$ given by weakly-order preserving maps. Or alternatively, a sequence of sets $X_0, X_1, X_2, \ldots$ with face maps $\partial^n_i : X_n \to X_{n-1}$ and degeneracy maps $\sigma^n_j : X_n \to X_{n+1}$ subject to certain identities $(0 \leq i, j, \leq n)$.

The category of simplicial sets is denoted **sSet**. It is clear **sSet** is the functor category $[\mathbf{\Delta}^{op}, \mathbf{Sets}]$.

Example 4.1.1. The category Δ[1] of standard n-simplices Δ^n with morphisms being the standard inclusion maps $\Delta^i \to \Delta^j$ for $0 \leq i \leq j$ gives a simplicial set by $\mathbf{n} \to \Delta^n$ and $f : \mathbf{i} \to \mathbf{j}$ associated to $\Delta^i \to \Delta^j$. In fact, this is an isomorphism of categories $\Phi : \mathbf{\Delta} \cong \Delta$.

Example 4.1.2. Any space X determines a simplicial set by the representable functor $\mathbf{Top}(\Phi(-), X)$. This construction is clearly a (covariant) functor $S : \mathbf{Top} \to \mathbf{sSet}$. It is called the *singular functor*.

A simplicial set X may be functorially turned into a topological space, and this functor is called Geometric Realization.

[1] not identical to $\mathbf{\Delta}$

Proposition 4.2 (Geometric Realization)**.** There exists a functor $|-|$: **sSet** $\to$ **Top** which is left-adjoint to the singular functor.

$$\mathbf{Top}(|X|, Y) \cong \mathbf{sSet}(X, SY)$$

naturally in X and Y.

We can define $|-|$ to be the functor taking X to the coend $\int_{\mathbf{n}\in\mathbf{\Delta}} X_n \times \Delta^n$ or equivalently, the colimit $\varinjlim_{\Delta^n \to X} \Delta^n$ in the slice category $\Delta \downarrow X$. The result follows due to Yoneda's Lemma. See [Maclane]. Furthermore, $|X|$ has a natural CW structure induced by CW structures on Δ^n.

Definition 4.3 (Nerve of a small category)**.** Let C be a small category. Then its nerve NC is a simplicial set defined by taking the n-simplices as functors $c : \mathbf{n} \to C$, i.e. diagrams of the form

$$c_0 \to c_1 \to \cdots \to c_{n-1}$$

for objects $c_0, c_1, \ldots, c_{n-1}$ in C. The i-th face map $\partial_i^n(c)$ on this simplex is obtained by deleting c_i in the evident way. Similarly, the j-th degeneracy map $\sigma_j^n(c)$ is obtained by replacing c_j by $c_j \to c_j$ in the obvious manner.

We define the *classifying space* of a small category C, denoted BC, as the geometric realization of NC.

Definition 4.4 (Classifying space of a small category)**.** $BC := |NC|$.

Example 4.1.3. For the categories $\mathbf{n}$, the geometric realization $B\mathbf{n} = \Phi\mathbf{n}$.

Furthermore, for any small category C, BC is the colimit (with the weak topology) of the directed system of the functors $B\mathbf{n} \to BC$ coming from $\mathbf{n} \to C$ and the morphisms coming from factorizations $\mathbf{i} \to \mathbf{n} \to C$.

Proposition 4.5. A functor $F : C \to D$ of small categories determines a cellular map $BF : BC \to BD$. Further, $B(\mathrm{id}_C) : BC \to BC$ is the identity map on BC, and for functors $F : C \to D, G : D \to E$, we have $B(G \circ F) = BG \circ BF$.

Proof. Due to the functoriality of geometric realization on simplicial sets, it suffices to show that the association $N : \mathbf{Cat} \to \mathbf{sSet}$ is functorial. But this follows directly from axioms of a category. □

Thus we have a functor $B : \mathbf{Cat} \to \mathbf{Top}$.

Example 4.1.4. A group G may be interpreted as a small category with a single object and all invertible arrows. We call BG the classifying space of the group G.

Proposition 4.6. The fundamental group $\pi_1(BG)$ is naturally isomorphic to G.

4.1.2 More Homotopy Theory

We will note that the following homotopy-theoretic generalization of a topological group will be relevant when considering upcoming constructions.

Definition 4.7 (H-space)**.** A H-space [2] X is a topological space with a binary operation $\mu : X \times X \to X$ and a distinguished element e such that

1. $\mu(e, e) = e$
2. $\mu(e, -) \simeq \mathrm{id}_X$
3. $\mu(-, e) \simeq \mathrm{id}_X$

Additionally, we say that a H-space is homotopy-associative or homotopy-commutative if those respective properties hold up to homotopy. A detailed description is available in [].

Topological groups are trivially examples of H-spaces. A class of illuminating nontrivial examples come from n-th loop spaces $\Omega^n X$ of some based space X, where the multiplication is given by concatenation of paths.

The (path) connected components of a homotopy-associative H-space determine a monoid.

We now describe the important notion of a *homotopy fiber*, which is useful for organizing how homotopy groups behave across a map $f : E \to B$.

Definition 4.8 (Homotopy Fiber)**.** Let $f : E \to B$ be a map of based spaces. The homotopy fiber $F(f)$ of this map is the homotopy-pullback of the cospan $E \to B \leftarrow PB$, where PB is the (based) space of paths in Y.

More concretely, we can define $F(f)$ to be consisting of pairs (e, γ) where $e \in E$ and γ is a path in B starting at the basepoint such that $f(e) = \gamma(1)$. $F(f)$ inherits the subspace topology from $E \times PB$ where PB itself carries the compact-open topology. We may declare $F(f)$ to have the basepoint $(*_E, *_{PB})$.

The key property of the homotopy fiber is that the sequence $F(f) \to E \to B$ of based space maps which yields a long exact sequence:

$$\cdots \to \Omega^{n+1} B \xrightarrow{\partial} \Omega^n F(f) \to \Omega^n E \to \Omega^n B \xrightarrow{\partial} \cdots$$
$$\cdots \xrightarrow{\partial} \Omega F(f) \to \Omega E \to \Omega B \xrightarrow{\partial} F(f) \to E \to B$$

which in turn yields a long exact sequence of groups/pointed sets:

$$\cdots \to \pi_{n+1} B \xrightarrow{\partial} \pi_n F(f) \to \pi_n E \to \pi_n B \xrightarrow{\partial} \cdots$$
$$\cdots \xrightarrow{\partial} \pi_1 F(f) \to \pi_1 E \to \pi_1 B \xrightarrow{\partial} \pi_0 F(f) \to \pi_0 E \to \pi_0 B$$

[2] The H stands for Hopf

Any other such sequence $F \to E \to B$ which results in the evident map $F \to F(f)$ (use $*_{PB}$ in the image) being a homotopy equivalence, is called a *homotopy fibration sequence*, and consequently also yields such a long exact sequence.

If $f : E \to B$ were a homotopy equivalence, then the homotopy fiber $F(f)$ is weakly contractible, i.e., it has the homotopy groups of a point, $\pi_* F(f) \cong \pi_*(\star)$. If E and B have homotopy type of CW complexes, then so is $F(f)$ and by Whitehead's Theorem [], weak homotopy equivalence implies homotopy equivalence.

Conversely, if the fiber $F(f)$ is contractible, then f is a weak homotopy equivalence, i.e., the induced maps $\pi_n(E) \to \pi_n(B)$ are isomorphisms. If E and B have homotopy type of CW complexes, then f is a homotopy equivalence.

4.1.3 +-construction on a connected CW complex

The +-construction, devised by ¡¿ in [], is a universal way to kill a perfect normal subgroup of the fundamental group of a connected CW complex, while preserving homology theory. The discussion of the +-construction fits into the framework of acyclic maps, which we explore here.

Definition 4.9. A space X is said to be acyclic if it has the homology of a point, i.e., $H_*(X; \mathbb{Z}) \cong H_*(\star; \mathbb{Z}) = \underline{\mathbb{Z}}$.

In particular, contractible spaces are acyclic. Acyclic spaces serve an analogous purpose to homology as contractible spaces do to homotopy.

Definition 4.10 (Acyclic maps)**.** Let $f : X \to Y$ be a map of based connected CW complexes. We say f is acyclic if the homotopy fiber $F(f)$ is acyclic, i.e., has the homology of a point.

In particular, a homotopy equivalence is acyclic since the former implies that the homotopy fiber is contractible. Note an acyclic map induces isomorphisms on integral homology $f_* : H_*(X; \mathbb{Z}) \cong H_*(Y; \mathbb{Z})$. Furthermore, we have the following result characterising acyclic maps:

Lemma 4.11. Let X, Y be based connected CW complexes. A map $f : X \to Y$ is acyclic if and only if for every $\pi_1(Y)$-module M, it induces an isomorphism $H_*(X; M) \cong H_*(Y; M)$.

The proof for this uses Serre Spectral sequences, which we skip and refer the reader to [Weibel].

Definition 4.12 (relative +-construction)**.** Let X be a based connected CW complex and P be a perfect normal subgroup of $\pi_1(X)$. Then we say a based acyclic map of CW-complexes $f : X \to Y$ is a +-construction relative to P if P is the kernel of the induced map $\pi_1(X) \to \pi_1(Y)$.

Since f was assumed to be acyclic, we can deduce from the exact sequence $\pi_1 F(f) \to \pi_1(X) \to \pi_1(Y) \to \pi_0 F(f)$ that the map $\pi_1(X) \to \pi_1(Y)$ is onto. (because $\pi_0 F(f) = \{\star\}$) And $\pi_1(Y) \cong \pi_1(X)/P$.

Theorem 4.13 (Quillen)**.** *Let X be a based connected CW complex and P be a perfect normal subgroup of $\pi_1(X)$.*

1. *There exists a +-construction $X \to Y$ relative to P.*
2. *A +-construction $f : X \to Y$ is universal among maps which kill P in the fundamental group. That is, for any $g : X \to Z$ such that the image of P in $\pi_1(Z)$ vanishes, we have a map $h : Y \to Z$, unique up to homotopy, such that $g = hf$.*
3. *In particular, if g is another +-construction relative to P, then there is a homotopy equivalence $h : Y \xrightarrow{\sim} Z$ compatible with the maps.*

The proof of this uses topological obstruction theory. We describe a construction for 1 in §4.1.4 and refer the reader to [Berrick] for a full proof of 1 and 2. 3 follows immediately from 2 since we have maps both way which must compose to identity up to homotopy.

Remark 4.1.1. Any group G has a maximal perfect normal subgroup, which we call the *perfect radical*. In fact, it is the subgroup generated by all perfect normal subgroups, hence their maximum. From now on, for any based CW complex X, we will denote by X^+ any +-construction on X relative to the perfect radical $P \triangleleft \pi_1(X)$.

4.1.4 Explicit Construction

A number of explicit *models* exist for the +-construction. Here, we present a construction due to [Kervaire] which constructs X^+ by attaching 2-cells to kill the maximal perfect subgroup of $\pi_1(X)$, and then 3-cells to recover the integral homology of X, making sure that the inclusion $X \hookrightarrow X^+$ is acyclic. We refer the reader to [Gersten] and [Suslin] for a variety of constructions. We will describe more models for K-theory in Sections §3 and §4.

Construction

Let X be a connected CW complex with a perfect normal subgroup P of the fundamental group $\pi_1(X)$. Identify the 1-cells in X corresponding to

of P as e_p^1 for each $p \in P$. Now construct a pushout Y by attaching disks D_p^2 with the boundary S_p^2 along every e_p^1. Call these new cells e_p^2.

That is, X' is the pushout of the diagram

$$\begin{array}{ccc} \coprod_{p\in P} S_p^1 & \hookrightarrow & \coprod_{p\in P} D_p^2 \\ \downarrow & & \downarrow \\ X & \hookrightarrow & Y \end{array}$$

Using the van-Kampen Theorem, we have the pushout through π_1 in **Grp**, which gives us $\pi_1(Y) = \pi_1(X) *_P 1 = \pi_1(X)/P$. (amalgamated product)

Consequently, H_1 is the abelianization. Since P is perfect, it is killed by abelianization and we have $H_1(Y) = H_1(X)$.

Consider the pair (Y, X). Since only 2-cells were attached, the homology of the pair is

$$H_2(Y, X) = \bigoplus_{[e_p^2]} \mathbb{Z}$$

and 0 elsewhere. We wish to cleverly attach 3-cells to kill relative homology without affecting H_3.

Consider the long exact sequence of the pair (Y, X)

$$\cdots \to H_{n+1}(Y, X) \xrightarrow{\partial} H_n(X) \to H_n(Y) \to H_n(Y, X) \xrightarrow{\partial} \cdots$$

At $n = 1$, since the map $H_1(X) \to H_1(Y)$ is an isomorphism, we have that $H_2(Y, X) \to H_1(X)$ is 0. Consequently, the map $H_2(Y) \to H_2(Y, X)$ is onto. We can lift each representative in $H_2(Y, X)$ to a representative in $H_2(Y)$, say σ_p for each class $[e_p^2]$.

Note each σ_p is represented by a map $h_p : S_p^2 \to Y$. Now attach 3-cells e_p^3 corresponding to each σ_p along the maps $h_p : S_p^2 \to Y$.

This yields another pushout which we call Z

$$Z = Y \cup \bigcup_{[e_p^2]} e_p^3$$

The triple $X \subseteq Y \subseteq Z$ induces a long exact sequence in relative homology

$$\cdots \to H_{n+1}(Z, Y) \xrightarrow{\partial} H_n(Y, X) \to H_n(Z, X) \to H_n(Z, Y) \xrightarrow{\partial} \cdots$$

At $n = 1$, since $H_2(Y, X) = 0$ and $H_0(Z, Y) = 0$, this gives us

$$\cdots \xrightarrow{\partial} 0 \to H_1(Z, X) \xrightarrow{\cong} H_1(Z, Y) \xrightarrow{\partial} 0 \to \cdots$$

Similarly, for $n \geq 4$, since $H_{n+1}(Z, Y) = 0$ and $H_{n-1}(Y, X) = 0$, we get

$$\cdots \xrightarrow{\partial} 0 \to H_n(Z, X) \xrightarrow{\cong} H_n(Z, Y) \xrightarrow{\partial} 0 \to \cdots$$

So we conclude $H_n(Z, X) \cong H_n(Z, Y) = 0$ for $n \neq 1, 2, 3$.

The only nontrivial terms in the sequence that remain are

$$\cdots \xrightarrow{\partial} 0 \to H_3(Z, X) \to H_3(Z, Y) \xrightarrow{\partial} H_2(Y, X) \to 0 \to \cdots$$

But $H_3(Z, Y)$ and $H_2(Y, X)$ are computable precisely due to construction. Z was constructed so as to kill

$$H_3(Z, Y) = \bigoplus_{[\sigma_p]} \mathbb{Z}$$

which is isomorphic via the evident maps to $H_2(Y, X) = \bigoplus_{[e_p]} \mathbb{Z}$.

So finally, we have $H_3(Z, X) = 0$, and altogether $H_n(Z, X) = 0$ for all n. Consequently, in the long exact sequence of the pair (Z, X), we get

$$\cdots \to H_{n+1}(Z, X) \xrightarrow{\partial} H_n(X) \to H_n(Z) \to H_n(Z, X) \xrightarrow{\partial} \cdots$$

which gives us that $H_n(X) \cong H_n(Z)$ induced via the inclusions are isomorphisms for all n.

It is clear that P is the kernel of the induced map $\pi_1(X) \to \pi_1(Y)$. This concludes the result that the stated construction preserves integral homology.

If this result were to extend to local coefficient systems, this would mean that the map $X \to Z$ is acyclic and forms a model for X^+. We do not pursue the full proof here and instead refer the reader to [Rosenberg]. Indeed, there is an alternate result in Proposition 4.19 which shows that what we have proven suffices for the case we are interested in, $BGL(R)^+$, which is what we shall discuss now.

4.2 The $BGL(R)^+$ construction for a ring R

?? We apply the +-construction on the classifying space $BGL(R)$ of the stable general linear group over R. The maximal perfect subgroup of $GL(R) = \pi_1 BGL(R)$ is the stable group of elementary matrices, $E(R)$.

Theorem 4.14. *With this definition of $BGL(R)^+$, we obtain:*

1. *$\pi_1 BGL(R)^+ \cong K_1(R)$ with the kernel of the induced map $GL(R) = \pi_1 BGL(R) \to \pi_1 BGL(R)^+$ being $E(R)$.*

2. *We have induced isomoprhisms* $H_*(BGL(R); M) \cong H_*(BGL(R)^+; M)$ *for every* $K_1(R)$*-module* M*.*

Proof. Since $E(R)$ is the maximal perfect normal subgroup of $GL(R)$, 1 follows immediately from Theorem 4.13. 2 follows from the characterization of acyclic maps in Lemma 4.11. □

Due to Theorem 4.13, $BGL^+(R)$ is uniquely defined up to homotopy equivalence via a canonical equivalence. Consequently, we may define $K_n(R) = \pi_n BGL(R)^+$ for $n \geq 1$. It is obvious that $K_1(R)$ defined here matches the description in Chapter 3. It remains to verify that this agrees with the classical theory for $n = 2$.

Proposition 4.15. Let P be a perfect normal subgroup of G with a $+$-construction $f : BG \to BG^+$ relative to P. Let $F(f)$ be the homotopy fiber of f. Then $\pi_1 F(f)$ is the universal central extension of P, and consequently $\pi_2(BG^+) \cong H_2(P; \mathbb{Z})$.

Proof. We have the fibration long exact sequence

$$\cdots \to \pi_2(BG) \to \pi_2(BG^+) \to \pi_1 F(f) \to G \to G/P \to 1$$

Since BG is an Eilenberg-Maclane space $K(G, 1)$, we have $\pi_2(BG) = 0$. So $\pi_2(BG^+)$ injects into $\pi_1 F(f)$, which in turns surjects onto the kernel of $G \to G/P$ which is P. Since $\pi_2(BG^+)$ is abelian, it must inject into the center of $\pi_1 F(f)$. This tells us that $\pi_1 F(f) \to P$ is a central extension. Since f is acylic, we have $\pi_1 F(f)$ is perfect and so $H_2(\pi_1 F(f), \mathbb{Z}) = 0$. By the Recognition Theorem [Chapter 3], we have that $\pi_1 F(f)$ is the universal central extension of P. Consequently, the kernel $\pi_2(BG^+)$ is given by $H_2(P; \mathbb{Z})$.

□

Corollary 4.16. We have $\pi_2(BGL(R))^+ \cong H_2(E(R); \mathbb{Z})$, and thus our definition matches with the classical theory since $H_2(E(R); \mathbb{Z}) \cong K_2(R)$ from Chapter 3.

However, at this point, it is not clear that the associations $R \mapsto BGL(R)^+$ or $R \mapsto K_n(R)$ are functorial. This invites some discussion.

4.2.1 Functoriality of the $BGL(R)^+$-construction

Apart from the upcoming models via $S^{-1}S$ and Q-constructions, we can make the following argument to show that the constructions of $BGL(R)^+$ can be performed functorially.

We can first define $BGL(\mathbb{Z})^+$ and take a homotopy pushout (using the mapping cylinder) via the canonical map $\mathbb{Z} \to R$ which induces $BGL(\mathbb{Z}) \to BGL(R)$. Note that the homotopy pushout

$$\begin{array}{ccc} BGL(\mathbb{Z}) & \longrightarrow & BGL(\mathbb{Z})^+ \\ \downarrow & & \downarrow \\ BGL(R) & \longrightarrow & BGL(R)^+ \end{array}$$

is a plus construction for $BGL(R)$.

4.2.2 H-Space Properties of $BGL(R)^+$

It's not hard to see that we can put a H-space structure on $BGL(R)^+$.

Proposition 4.17. $BGL(R)^+$ is a homotopy-associative H-space.

Proof. We have the group map $\oplus : GL_m(R) \times GL_n(R) \to GL_{m+n}(R) \subset GL(R)$ which induces $BGL_m(R) \times BGL_n(R) \to BGL(R)$. Since we have show that the plus construction can be taken to be functorial, we have maps

$$(BGL_m(R) \times BGL_n(R))^+ \to BGL(R)^+$$

. [Finish this] □

There is a recognition criteria for +-constructions of $BGL(R)$ in terms of homology and H-spaces, due to Quillen. The proof again uses obstruction theory. See [Gersten].

Theorem 4.18. *The plus construction $i : BGL(R) \to BGL(R)^+$ is universal for maps into H-spaces. That is, for any map $f : BGL(R) \to H$ for some H-space H, we have a map $g : BGL(R)^+ \to H$ so that $f = gi$, such that g is unique up to weak homotopy, i.e. the induced maps $\pi_n(BGL(R)^+) \to \pi_n(H)$ is independent of g.*

Proposition 4.19. If further $f_* : H_*(BGL(R); \mathbb{Z}) \xrightarrow{\cong} H_*(H; \mathbb{Z})$ is an isomorphism, then f is acyclic and g is a homotopy equivance.

Proof. It suffices to prove that f is acyclic. □

4.3 $S^{-1}S$ construction

For a symmetric monoidal category $(S, \square)$, the construction of Chapter 2 for K_0 may be extended to a K-theory space $K^{\square}(S)$ by constructing an

appropriate group-completion of S with respect to the monoidal operation $\square$.

Note that the geometric realization of a symmetric monoidal category is a homotopy-associative homotopy-commutative H-space with the multiplication coming from the monoidal product $\square$. This is because [] We wish to construct a symmetric monoidal category $S^{-1}S$ which describes a "group completion" of BS so as to make $BS^{-1}S$ be a group-like H-space. [Grayson]

But first, for either of BS or $BS^{-1}S$ to be interesting, we must reduce our scope.

Definition 4.20 (symmetric monoidal groupoid)**.** A *symmetric monoidal groupoid* is a symmetric monoidal category $(S, \square)$ in which all morphisms are isomorphisms.

Further, we say S is *faithful* if all translations $s\square : \mathrm{Aut}(t) \to \mathrm{Aut}(s\square t)$ are injective.

Definition 4.21. Given a symmetric monoidal category S, $\mathrm{iso}(S)$ is the symmetric monoidal groupoid which is a subcategory of S consisting of the same objects but inheriting only isomorphisms.

A variety of symmetric monoidal groupoids yield interesting geometric realizations.

Example 4.3.1. The category $\mathbf{F}(R)$ with $\square = \oplus$ is contractible since it has an initial object 0. However $\mathrm{iso}(\mathbf{F}(R))$ is more interesting and in fact equivalent to the disjoint union $\coprod_n GL_n(R)$ where n runs through the integers.

Example 4.3.2. Similarly, 0 is also an initial object for the category $\mathbf{P}(R)$ with $\square = \oplus$. But $\mathrm{iso}(\mathbf{P}(R))$ is equivalent to $\coprod_P \mathrm{Aut}(P)$ where P runs over representatives of isomorphism clases of finitely generated projective modules.

Let S be a faithful symmetric monoidal groupoid.

Definition 4.22 ($S^{-1}S$)**.** Define $S^{-1}S$ to be a category where the objects are pairs of objects (m, n) for $m, n \in S$. Morphisms in $S^{-1}S$ from $(m_1, m_2) \to (n_1, n_2)$ are equivalence classes of triples (s, f_1, f_2) for some object $s \in S$ and $(f_1, f_2) : (s\square m_1, s\square m_2) \to (n_1, n_2)$ being a morphism in $S \times S$.

That is,

$$(m_1, m_2) \xrightarrow{s\square} (s\square m_1, s\square m_2) \xrightarrow{(f_1, f_2)} (n_1, n_2)$$

which we consider equivalent to

$$(m_1, m_2) \xrightarrow{t\square} (t\square m_1, t\square m_2) \xrightarrow{(g_1, g_2)} (n_1, n_2)$$

if there is an isomorphism $\alpha : s \xrightarrow{\cong} t$ such that $\alpha \square g_1 = f_1$ and $\alpha \square g_2 = f_2$. The composition $[(s, f_1, f_2)] \circ [(t, g_1, g_2)]$ is defined as $[(s \square t, f_1 \circ g_1, f_2 \circ g_2)]$. One may verify this is well-defined and indeed forms a category.

Remark 4.3.1. Consider two distinguished classes of morphisms in $S^{-1}S$. One is simultaneous translations, which we formally write $s\square : (m_1, m_2) \to (s\square m_1, s\square m_2)$, represented by $(s, \mathrm{id}, \mathrm{id})$. The other being pairs of maps $(f_1, f_2) : (m_1, m_2) \to (n_1, n_2)$ coming from morphisms in $S \times S$ where the translation composite is trivial. That is, the morphism is represented by (e, f_1, f_2).

Proposition 4.23. Let S be a faithful symmetric monoidal groupoid. Then any morphism $[(s, f_1, f_2)]$ in $S^{-1}S$ determines s, f_1, f_2 uniquely up to unique isomorphisms.

Proof. It is clear s, f_1, f_2 are determined uniquely up to isomorphism. Suppose (s, f_1, f_2) is equivalent to (t, g_1, g_2) via isomorphisms $\alpha, \beta : s \xrightarrow{\cong} t$ Since $\alpha \square g_i = f_i = \beta \square g_i$, we have $\alpha \square m_i = \beta \square m_i$ and since S was faithful, we recover $\alpha = \beta$. □

Thus, we can think of morphisms $[(s, f_1, f_2)]$ in $S^{-1}S$ as comprising of a *translation component* $[s\square]$ and *mapping component* $[(f_1, f_2)]$.

Remark 4.3.2. $S^{-1}S$ is a symmetric monoidal category with the multiplication given by $(m_1, m_2)\square(n_1, n_2) := (m_1 \square n_1, m_2 \square n_2)$. It is clear that the functor $S \to S^{-1}S$, which sends $s \mapsto (s, e)$ is a monoidal functor. Further, a monoidal functor $S \to T$ describes a (monoidal) natural transformation

$$\begin{array}{ccc} S & \longrightarrow & T \\ \downarrow & & \downarrow \\ S^{-1}S & \longrightarrow & T^{-1}T \end{array}$$

We also have an induced H-space map $BS \to B(S^{-1}S)$, and a map $\pi_0(S) \to \pi_0(S^{-1}S)$ of abelian monoids. (The component of s maps to the component of (s, e))

Further, $\pi_0(S^{-1}S)$ is an abelian group since for the connected coponent of (m, n) we have we have inverse (n, m) due to the isomorphism $(e, e) \xrightarrow{\cong} (m, n)\square(n, m)$ given by the mapping component $[(c_{mn}, c_{mn}^{-1})]$ where c_{mn} is the symmetric braiding given by isomorphism $m\square n \to n\square m$ in S. It is abelian due to S being symmetric.

Definition 4.24. Let S be a symmetric monoidal groupoid. We define

$$K^{\square}(S) := B(S^{-1}S)$$

and consequently we can define $K_n^{\square}(S) = \pi_n B(S^{-1}S)$.

Note that a (strict) monoidal functor $S \to T$ induces $S^{-1}S \to T^{-1}T$ which in turn induces $K^{\square}(S) \to K^{\square}(T)$, Thus the $S^{-1}S$ construction is functorial!

Proposition 4.25. The group $K_0^{\square}(S) = \pi_0(S^{-1}S)$ is indeed the group completion of the monoid given by isomorphism classes of S via the inclusion $i_* : \pi_0(S) \to \pi_0(S^{-1}S)$.

Proof. Let $\varphi : \pi_0(S) \to G$ be a group completion of $\pi_0(S)$. We will show that the universal homomorphism $\widetilde{i}_* : G \to \pi_0(S^{-1}S)$ is invertible. Let j be the set map $S^{-1}S \to G$ as $j(m,n) := \varphi(m) - \varphi(n)$. If we have $s \in S$ then $j(m_1, m_2) = j(s\square m_1, s\square m_2)$ and if $f_i : m_i \to n_i$, then $j(m_1, m_2) = \varphi(m_1) - \varphi(m_2) = \varphi(n_1) - \varphi(n_2) = j(n_1, n_2)$. This shows j induces a well-defined homomorphism $j_* : \pi_0(S^{-1}S) \to G$. By construction, it is the inverse to $G \to \pi_0(S^{-1}S)$. □

This shows our definition matches the definition of $K_0(S)$ in Chapter 2.

4.3.1 Group Completions of BS

We turn our attention to

Theorem 4.26. *If S*

Theorem 4.27 (Quillen)**.** *Let $S = \coprod GL_n(R)$ which is equivalent to $\mathbf{F}(R)$. Then $B(S^{-1}S)$ is the group-completion of BS and*

$$B(S^{-1}S) \simeq \mathbb{Z} \times BGL(R)^+$$

Proof. □

Theorem 4.28. *Let S be a symmetric monoidal groupoid with faithful transitions satisfying cofinality.*

We have then

$$K^{\square}(S) \simeq K_0^{\square}(S) \times B\,\mathrm{Aut}(S)^+ A$$

4.4 Quillen's Q-construction

In [Quillen, 19XX], Quillen proposed another construction for the K-theory space. Given an exact category $\mathcal{A}$, [] we will define the category $Q\mathcal{A}$, and then define $K_n(\mathcal{A}) = \pi_{n+1}(BQ\mathcal{A})$ for all $n \geq 0$.

Definition 4.29 (Q-construction)**.** Let $\mathcal{A}$ be an exact category. We define $Q\mathcal{A}$ as the category with the same objects as $\mathcal{A}$, and the morphisms between $C \to C'$ are equivalence classes of diagrams

$$C \xleftarrow[j]{} D \xrightarrow{i} C'$$

where two such diagrams are considered isomorphic when there is an isomorphism of these diagrams which is identity on C and C'.

The composition of the above arrow with $C' \twoheadleftarrow D' \rightarrowtail C''$ is given by $C \twoheadleftarrow D \times_{C'} D' \rightarrowtail C''$

$$\begin{array}{ccccc} D \times_{C'} D' & \overset{\pi_2}{\rightarrowtail} & D' & \overset{i'}{\rightarrowtail} & C'' \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle j'} & & \\ D & \underset{i}{\rightarrowtail} & C' & & \\ {\scriptstyle j}\downarrow & & & & \\ C & & & & \end{array}$$

It is clear that composition is well-defined and associative and so $Q\mathcal{A}$ forms a well-defined category.

[Set theoretic remark]

Remark 4.4.1. Admissible monomorphisms $C \rightarrowtail C'$ determine a morphism $C \to C'$ by taking $D = C'$ and $D \twoheadleftarrow C'$ as identity. We will call these morphisms *injective*.

Similarly, admissible epimorphisms $C \twoheadleftarrow C'$ determine a morphism $C \to C'$ by taking $D = C$ and $C \rightarrowtail D$ as identity. We will call these morphisms *surjective*.

Note that these distinguished families of morphisms are closed under composition, and generate all morphisms. Furthermore, any morphism $C \twoheadleftarrow D \rightarrowtail C'$ factors as $i_* j^*$ where i_* is the injective morphism determined by $D \rightarrowtail C'$ and j^* is the surjective morphism determined by $C \twoheadleftarrow D$. This factorization is unique up to unique isomorphism.

Alternative Description of Morphisms in $Q\mathcal{A}$

There is an alternative way to describe the morphisms $C \to D$ in $Q\mathcal{A}$ in terms of *subobjects* and *subquotients*. Fix some object C in $\mathcal{A}$.

Definition 4.30. An *admissible subobject* of C is an equivalence class of admissible monomorphisms $C' \rightarrowtail C$ with equivalence given by whether they factor through each other, or in other words as isomorphism in $\mathcal{A} \downarrow C$.

An *admissible subquotient* is defined in the dual manner, using admissible epimorphisms.

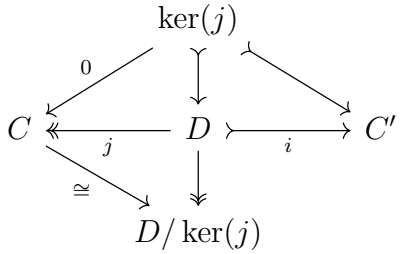
One can define a partial order on the set of admissible monomorphisms into C. We declare $C_1 \leq C_2$ if there is an admissible monomorphism $C_1 \rightarrowtail C_2$,

which if it exists has to be unique. We then say (C_1, C_2) is an *admissible layer* in C and the cokernel C_2/C_1 is an *admissible subquotient* of C.

Proposition 4.31. The morphisms from $C \to C'$ may be identified with pairs $((C_1, C_2), \varphi)$, where (C_1, C_2) is an admissible layer in C' and $\varphi : C \xrightarrow{\cong} C_2/C_1$ is an isomorphism.

Furthermore, composition of morphisms $C \to C'$ and $C' \to C''$ is associative and arises from the fact that subquotient of a subquotient is a subquotient.

Proof. For $C \twoheadleftarrow D \rightarrowtail C'$, take the admissible layer (C_1, C_2) where $C_1 = D$ and C_2 is the kernel of $C \twoheadleftarrow D$. Clearly, since C_2 gives a subobject of $D = C_1$ which gives a subobject of C', the given pair is indeed an admissible layer. The isomorphism follows since both C and the C_2/C_1 are cokernels of the kernel of an admissible epimorphism.



Conversely, an admissible layer (C_1, C_2) in C evidently determines a morphism $C_2/C_1 \twoheadleftarrow C_2 \rightarrowtail C$ in $Q\mathcal{A}$. □

Proposition 4.32. An exact functor $F : \mathcal{A} \to \mathcal{B}$ induces a functor $QF : Q\mathcal{A} \to Q\mathcal{B}$.

Theorem 4.33 (Quillen)**.** *The fundmental group of $BQ\mathcal{A}$ at the basepoint 0 is canonically isomorphic to the $K_0(\mathcal{A})$ group defined in Chapter 2.*

Proof. □

[Set Theory remark]

Definition 4.34. Let $\mathcal{A}$ be a small exact category. Then $K\mathcal{A}$ is defined to be the space $\Omega BQ\mathcal{A}$ and consequently, we define

$$K_n(\mathcal{A}) := \pi_n(K\mathcal{A}) = \pi_{n+1}(BQ\mathcal{A})$$

[Functoriality]

Example 4.4.1. If $\mathcal{A}$ is a small category equivalent to the category $\mathbf{P}(R)$ of finitely generated projective modules over R, then we define $K_n(R) := K_n(\mathcal{A})$. It is clear this matches our original definition for $n = 0$. We prove in §4.5 the $+ = Q$ Theorem, which implies that this matches with the description of the K_n groups defined using the +-construction in §**??**.

4.4.1 Key Properties of the Q-construction

4.5 Quillen's $+ = Q$ Theorem

Given an additive category $\mathcal{A}$, there are two approaches to defining K-theory over $\mathcal{A}$.

Theorem 4.35 (Quillen)**.** *Let $\mathcal{A}$ be a split-exact category and $S = \operatorname{iso}\mathcal{A}$, then*

$$\Omega BQ\mathcal{A} \simeq B(S^{-1}S)$$

Corollary 4.36 (+=Q)**.** For any unital ring R,

$$\Omega BQ\mathbf{P}(R) \simeq K_0(R) \times BGL(R)^+$$